

Power Query Data Cleaning & Automation Workflows

1. Import CSV File As Table In Excel

The image shows a CSV file named 'salesdata.csv' with the following data:

OrderID	Date	Customer	Product	Quantity	Revenue
1001	2026-05-01	Sarah	Laptop	1	950
1002	2026-05-01	James	Mouse	2	40
1003	2026-05-02	Priya	Keyboard	1	75
1004	2026-05-02	Ahmed	Monitor	2	400
1005	2026-05-03	Lucy	Headphones	1	120
1006	2026-05-04	Emma	Webcam	1	85

Below the CSV content, the 'Import Data' dialog box is shown. It has the following settings:

- File Origin: 1252: Western European (Windows)
- Delimiter: Comma
- Data Type Detection: Based on first 200 rows

The 'Load' button is highlighted in green.

Below the dialog box, the Power Query 'Query Settings' pane is visible. It shows the query name 'salesdata' and the applied steps: 'Source', 'Promoted Headers', and 'Changed Type'.

Here we import a CSV File of comma separated values into an excel spreadsheet. The reason this is often done is because its often information we wish to use as part of a dataset or it can represent new records to add to our existing data set.

2. Cleaning Messy Data

```
messydata.csv
File Edit View

OrderID,Date,Customer,Product,Quantity,Revenue
1001,2026-05-01,Sarah ,Laptop,1,950
1002,2026-05-01,,Mouse,2,40
1003,not entered,Priya,Keyboard,1,75
1004,2026-05-02,Ahmed,Monitor,,400
1004,2026-05-02,Ahmed,Monitor,,400
1005,2026-05-03, Lucy,Headphones,1,120
```

messydata.csv

File Origin

1252: Western European (Windows)

Delimiter

Comma

Data Type Detection

Based on first 200 rows

OrderID	Date	Customer	Product	Quantity	Revenue
1001	2026-05-01	Sarah	Laptop	1	950
1002	2026-05-01		Mouse	2	40
1003	not entered	Priya	Keyboard	1	75
1004	2026-05-02	Ahmed	Monitor	null	400
1004	2026-05-02	Ahmed	Monitor	null	400
1005	2026-05-03	Lucy	Headphones	1	120

Load

Transform Data

Cancel

Before

Manage Columns

Remove Top Rows

Remove Bottom Rows

Remove Alternate Rows

Remove Duplicates

Remove Blank Rows

Remove Errors

Transform

Combine

Parameters

Data Sources

New Query

Table

OrderID

1

2

3

4

5

6

2026-05-01

2026-05-01

not entered

2026-05-02

2026-05-02

2026-05-03

Sarah

Priya

Ahmed

Ahmed

Lucy

Laptop

Mouse

Keyboard

Monitor

Monitor

Headphones

1

2

1

null

null

1

950

40

75

400

400

120

Qu

PR

Nai

m

All

AP

After

	OrderID	Date	Customer	Product	Quantity	Revenue
1	1001	2026-05-01	Sarah	Laptop	1	950
2	1002	2026-05-01		Mouse	2	40
3	1003	not entered	Priya	Keyboard	1	75
4	1004	2026-05-02	Ahmed	Monitor	null	400
5	1005	2026-05-03	Lucy	Headphones	1	120

Format Date Column To Of Type Date

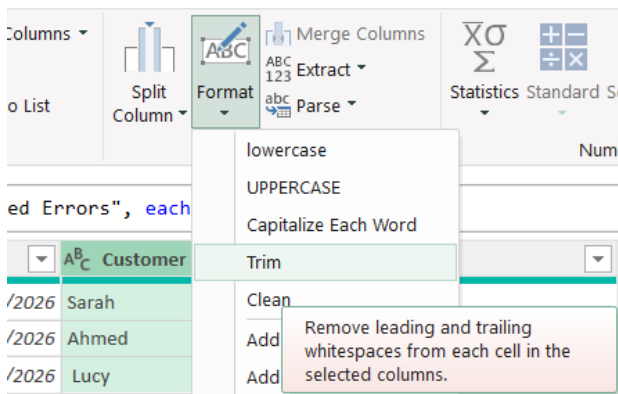
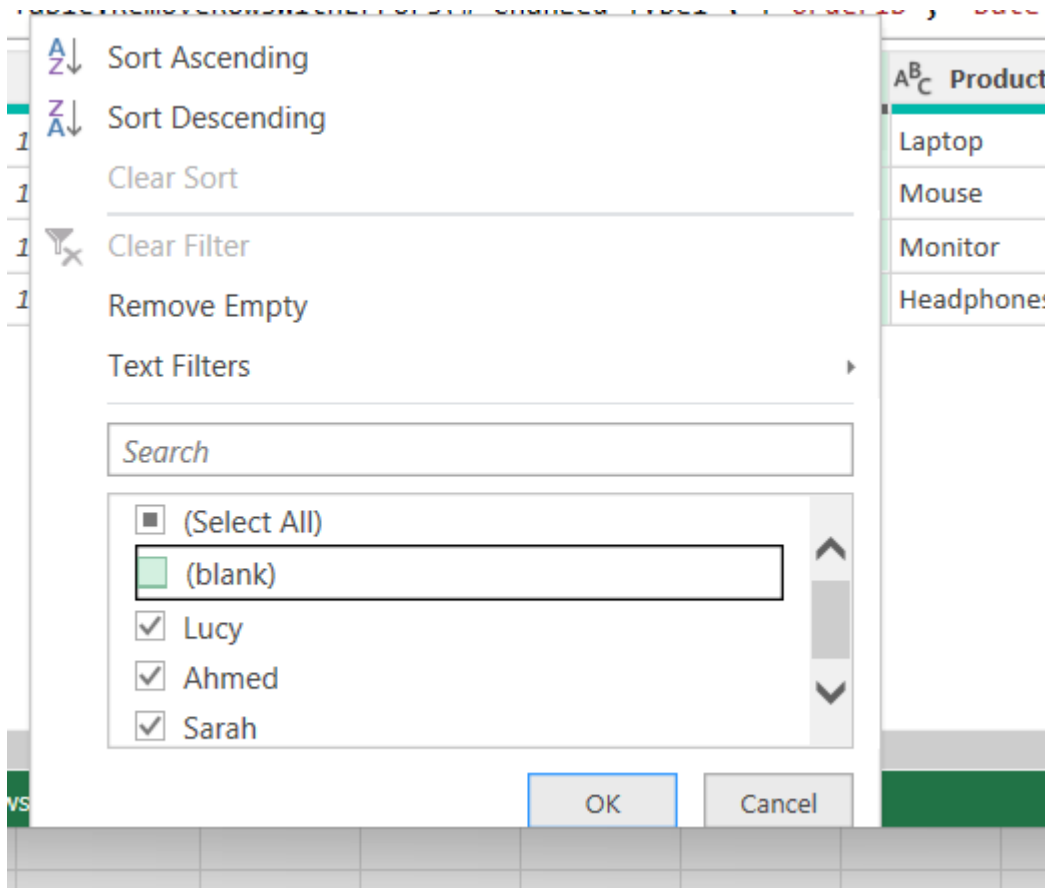
	OrderID	Date	Customer	Product	Quantity	Revenue
1	1001	Decimal Number	Sarah	Laptop	1	950
2	1002	Currency		Mouse	2	40
3	1003	Whole Number	Priya	Keyboard	1	75
4	1004	Percentage	Ahmed	Monitor	null	400
5	1005	Date/Time	Lucy	Headphones	1	120

	OrderID	Date	Customer	Product	Quantity	Revenue
1	1001	01/05/2026	Sarah	Laptop	1	950
2	1002	01/05/2026		Mouse	2	40
3	1003	Error	Priya	Keyboard	1	75
4	1004	02/05/2026	Ahmed	Monitor	null	400
5	1005	03/05/2026	Lucy	Headphones	1	120

	OrderID	Date	Customer	Product	Quantity	Revenue
1	1001	01/05/2026	Sarah	Laptop	1	950
2	1002	01/05/2026		Mouse	2	40
3	1003	Error	Priya	Keyboard	1	75
4	1004	02/05/2026	Ahmed	Monitor	null	400
5	1005	03/05/2026	Lucy	Headphones	1	120

	OrderID	Date	Customer	Product	Quantity	Revenue
1	1001	01/05/2026	Sarah	Laptop	1	950
2	1002	01/05/2026		Mouse	2	40
3	1004	02/05/2026	Ahmed	Monitor	null	400
4	1005	03/05/2026	Lucy	Headphones	1	120

	OrderID	Date	Customer	Product	Quantity	Revenue
1	1001	01/05/2026	Sarah	Laptop	1	950
2	1002	01/05/2026		Mouse	2	40
3	1004	02/05/2026	Ahmed	Monitor	null	400
4	1005	03/05/2026	Lucy	Headphones	1	120



	OrderID	Date	Customer	Product	Quantity	Revenue
1	1001	01/05/2026	Sarah	Laptop	1	950
2	1004	02/05/2026	Ahmed	Monitor	null	400
3	1005	03/05/2026	Lucy	Headphones	1	120

1 ² ₃ Quantity	1 ² ₃ Revenue
1	950
	400
	120

- Copy
- Number Filters
- 1₂ Replace Values...
- Drill Down
- Add as New Query

1 ² ₃ Quantity	1 ² ₃ Revenue
1	950
null	400
1	120

Replace Values

Replace one value with another in the selected columns.

Value To Find

Replace With

OK

Cancel

OrderID	Date	A ^B _C Customer	A ^B _C Product	1.2 Quantity	1 ² ₃ Revenue
1	1001	01/05/2026 Sarah	Laptop	1	950
2	1004	02/05/2026 Ahmed	Monitor	1	400
3	1005	03/05/2026 Lucy	Headphones	1	120

	OrderID	Date	Customer	Product	Quantity	Revenue								
2	1001	01/05/2026	Sarah	Laptop	1	950								
3	1004	02/05/2026	Ahmed	Monitor	1	400								
4	1005	03/05/2026	Lucy	Headphones	1	120								
5														
6														
7														
8														

Here we perform multiple operations on our CSV file such as removing duplicates, errors and replacing values to make our dataset cleaner.

3. Merge In Power Query

Merge

Select a table and matching columns to create a merged table.

sales

OrderID	CustomerID	Revenue	CustomerName	Region
1001	C001	950	Sarah	London
1002	C002	40	James	Manchester
1003	C003	75	Priya	Birmingham

Refresh

No preview is available

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

Fuzzy matching options

OK

Cancel

Merge

Select a table and matching columns to create a merged table.

sales

OrderID	CustomerID	Revenue
1001	C001	950
1002	C002	40
1003	C003	75

customers

CustomerID	CustomerName	Region
C001	Sarah	London
C002	James	Manchester
C003	Priya	Birmingham

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

▸ Fuzzy matching options

OK

Cancel

Merge

Select tables and matching columns to create a merged table.

sales

OrderID	CustomerID	Revenue
1001	C001	950
1002	C002	40
1003	C003	75

customers

CustomerID	CustomerName	Region
C001	Sarah	London
C002	James	Manchester
C003	Priya	Birmingham

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

▸ Fuzzy matching options

✓ The selection matches 3 of 3 rows from the first table.

OK

Cancel

	AB	CustomerID	Revenue	customers
1	1	C001	950	Table
2	2	C002	40	Table
3	3	C003	75	Table

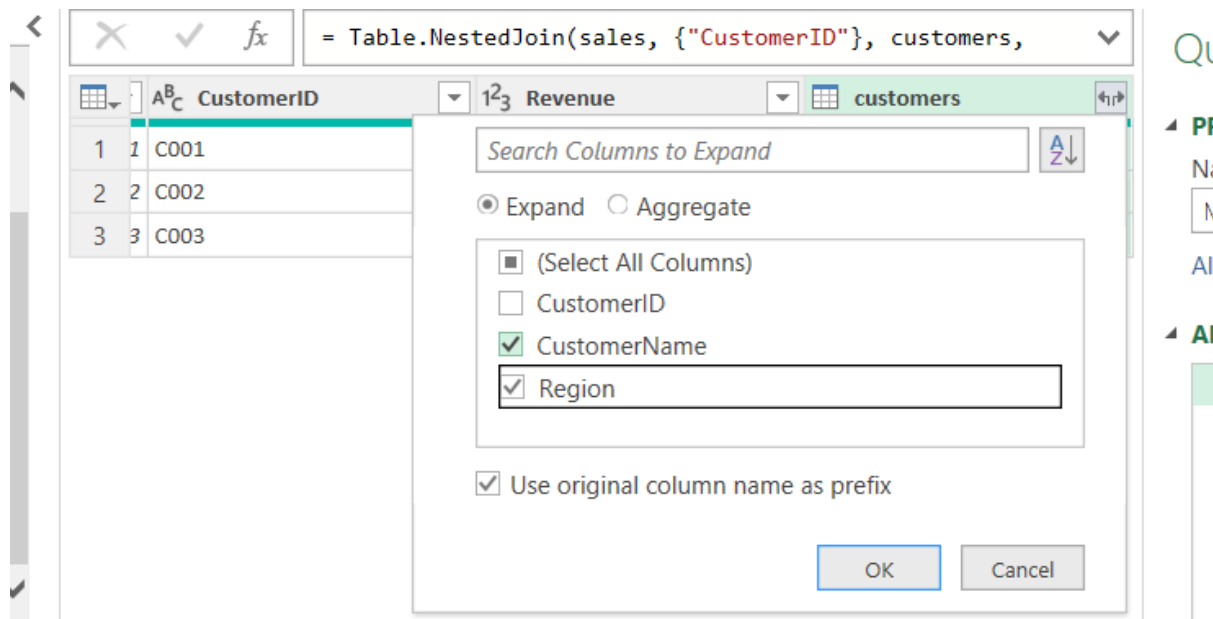
Query Setting:

PROPERTIES

Name

Merge1

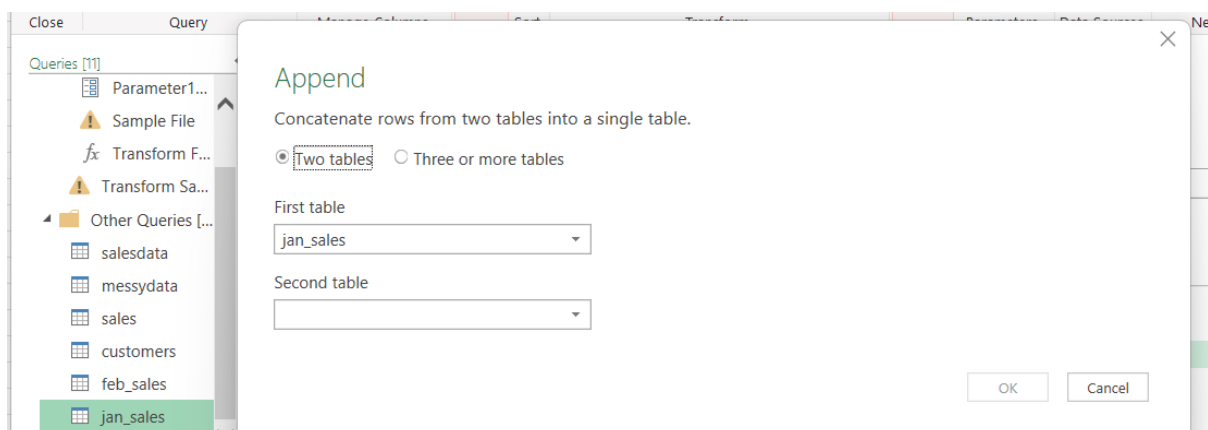
All Properties

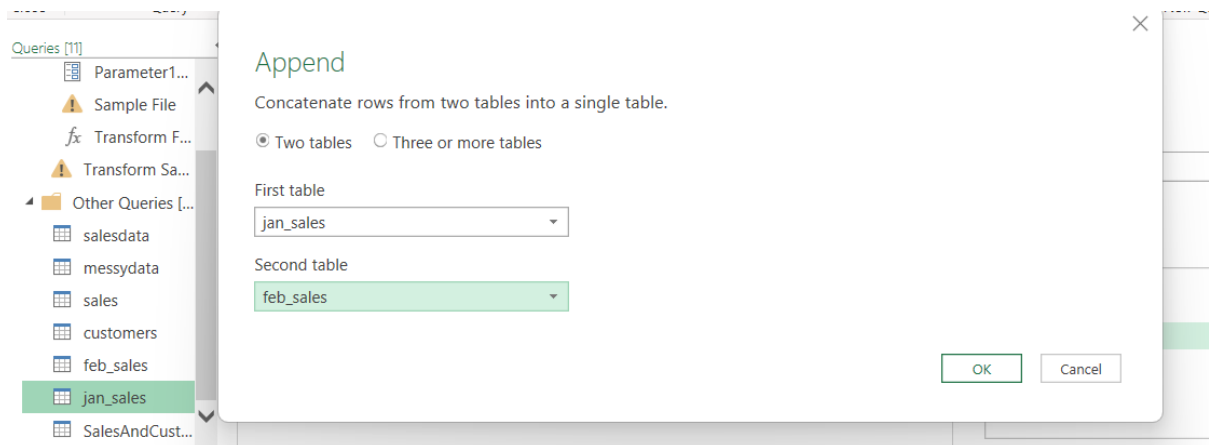


OrderID	CustomerID	Revenue	customers.CustomerName	customers.Region
1001	C001	950	Sarah	London
1002	C002	40	James	Manchester
1003	C003	75	Priya	Birmingham

Merge In Excel gives you the ability to append extra columns to a dataset given 2 tables contain a matching column which in this case is CustomerId.

4. Append In Power Query

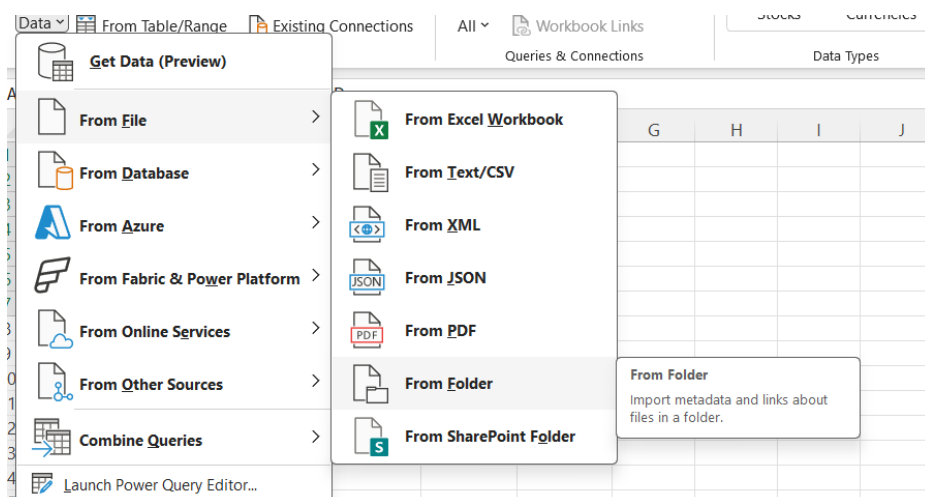




OrderID	Revenue
1001	950
1002	40
1003	75
1004	400

Append Query provides you with the ability to combine csv's into one table like this in example we can combine January and February Sales.

5. Folder Based Automation



C:\Users\user\OneDrive\Documents\WeeklySales

Content	Name	Extension	Date accessed	Date modified	Date created	Attributes	Folder Path
Binary	sales_week1.csv	.csv	10/05/2026 14:44:14	10/05/2026 13:57:02	10/05/2026 14:44:13	Record	C:\Users\user\OneDrive\Docume
Binary	sales_week2.csv	.csv	10/05/2026 14:44:14	10/05/2026 13:57:35	10/05/2026 14:44:13	Record	C:\Users\user\OneDrive\Docume

Combine

Load

Transform Data

Cancel

Combine Files

Specify the settings for each file. [Learn more](#)

Sample File:

sales_week2.csv

File Origin

1252: Western European (Windows)

Delimiter

Comma

Data Type Detection

Based on first 200 rows

OrderID	Revenue
1003	75
1004	400

☐ Skip files with errors

OK

Cancel

= Table.TransformColumnTypes("#Expanded Table Column1",			
	Source.Name	OrderID	Revenue
1	sales_week1.csv	1001	950
2	sales_week1.csv	1002	40
3	sales_week2.csv	1003	75
4	sales_week2.csv	1004	400

	A	B	C
1	Source.Name	OrderID	Revenue
2	sales_week1.csv	1001	950
3	sales_week1.csv	1002	40
4	sales_week2.csv	1003	75
5	sales_week2.csv	1004	400
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			
17			
18			
19			
20			
21			
22			
23			
24			
25			
26			
27			
28			

WeeklySales (2) salesdata

With Folder Based Automation new weekly sales files can be dropped into the folder and automatically ingested via refresh without rebuilding transformations.